

Homework

Example (Con.) WH Location problem

Decision variable $\mathbf{v} = \begin{bmatrix} x \\ y \end{bmatrix}$ location of the warehouse

Euclidean distance (ℓ_2 norm)

Minimize $\sum_{j=1}^4 r_j \|\mathbf{u}_j - \mathbf{v}\|_2 = \sum_{j=1}^4 r_j \sqrt{(x_j - x)^2 + (y_j - y)^2}$

Rectilinear distance (ℓ_1 norm)

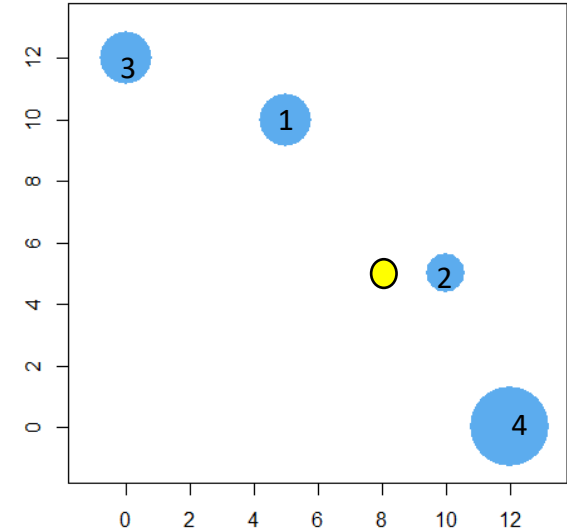
$$\min_{x,y} f(x,y) = \sum_{j=1}^4 r_j \|\mathbf{u}_j - \mathbf{v}\|_1 = \sum_{j=1}^4 r_j (|x_j - x| + |y_j - y|)$$

To linearize, we introduce auxiliary variables s_j and t_j

$$\begin{aligned} \min_{x,y,s_j,t_j} \quad & \sum_j r_j (s_j + t_j) \\ & s_j \geq x_j - x, \quad s_j \geq -(x_j - x) \quad \forall j \\ & t_j \geq y_j - y, \quad t_j \geq -(y_j - y) \quad \forall j \end{aligned}$$

Revisit Slide05: Euclidean distance

Customer	Customer location $\mathbf{u}_j = \begin{bmatrix} x_j \\ y_j \end{bmatrix}$		Distance $\ \mathbf{u}_j - \mathbf{v}\ _2$	# of shipments per year r_j
j	Lat	Long		
1	5	10	6.58	200
2	10	5	0.69	150
3	$x_3 = 0$	$y_3 = 12$	11.63	200
4	12	0	5.70	300
WH location			$x_0 = 9.31$ $y_0 = 5.03$	
Total yearly distance from WH to customers				5,456.54



We want to find the warehouse location (x,y) that minimizes the total yearly distance. Here, we use the *rectilinear* distance instead of the Euclidean distance.

$$\min_{x,y} 200(|5 - x| + |10 - y|) + \dots + 300(|12 - x| + |0 - y|)$$

1. Linearize the problem.
2. Solve for the optimal warehouse location.

Homework (optional): Uncapacitated Facility

Location (UFL)

Inputs

$M = \{1, \dots, m\}$ set of clients

$N = \{1, \dots, n\}$ set of potential depots

f_j = fixed cost of locating at candidate site j (THB)

d_{ij} = distance from demand node i to candidate location j (km)

h_i = demand at node i (kg)

α = cost per unit distance per unit demand (THB/km/kg) := 0.35 THB/kg/km

$c_{ij} = \alpha h_i d_{ij}$ cost if all of client i 's order is delivered from depot j

c_{LK} = cost if all of client L order is delivered from depot K

$c_{LK} = \alpha h_L d_{LK} = (0.35 \text{ THB/kg/km})(20 \text{ kg})(21 \text{ km}) = (7 \text{ THB/km})(21 \text{ km}) = 147 \text{ THB}$

Decision variables

x_{ij} = fraction of demand of client i satisfied from depot j

$y_j = 1$ if depot j is used, and $y_j = 0$ o/w

$$\min \sum_{i \in M} \sum_{j \in N} c_{ij} x_{ij} + \sum_{j \in N} f_j y_j$$

[satisfaction of demand of client i]

[the link between x_{ij} and y_j variables]

$$\sum_{j \in N} x_{ij} = 1 \quad \forall i \in M$$

$$x_{ij} \leq y_j \quad \forall i \in M, j \in N$$

Solve this problem.

$$x_{ij} \geq 0, \quad \forall i \in M, j \in N$$

$$y_j \in \{0,1\} \quad \forall j \in N$$

